Statewide Bus-Level Load Reconstruction

Results Summary for the Paper

Rafaël Leclerc

October 2026

Scope. This is the condensed, paper-facing version: the numbers and figures needed to write, with no methodology history. The full record — every bug found, every intermediate pipeline version, every audit — is in the companion `report.pdf` (33 pp.), referenced here only where a claim needs its provenance.

What changed since the Coast-only study. The framework now runs on the full Texas 7k grid: **6,717 buses**, five IECC climate zones, residential and commercial, validated against **all eight ERCOT weather zones**. Three structural limitations of the Coast study were removed: the winner-takes-all spatial join, the single-zone validation, and the hand-picked industrial constant.

1 Method, in brief

Load models. Per IECC climate zone, LightGBM models map weather and calendar features to normalized load: 5 ResStock residential archetypes (kWh per dwelling) and 15 ComStock commercial archetypes (kWh per ft^2). Features are the four weather channels (temperature, relative humidity, wind speed, GHI), cyclical hour/month/day-of-week, and lagged/rolling temperature (1 h, 3 h, 24 h mean) — the last group being the single largest accuracy gain found in the Coast study. Weather is a housing-unit-weighted blend of the NREL AMY-2018 county files across each zone's counties. Accuracy is reported under **blocked seasonal 4-fold cross-validation** (train on three seasons, test on the held-out one); production models are then trained on the full year.

Spatial allocation. Each 2020 census tract is subdivided into 50 equal-area parts and each 2010 ZCTA into 100, and every part is joined to its nearest bus. The allocation weight is

$$w(\text{polygon}, \text{bus}) \;=\; \frac{\text{parts won by that bus}}{\text{total parts}},$$

so a polygon's dwellings and floor space are spread across every bus serving part of it. Weights sum to one per polygon, so the allocation conserves **100.00% of Texas housing units** (11,628,419). Bus-level counts come from Census B25024 (dwellings by structure type, 2020 tract vintage) and from County Business Patterns employment converted to floor space via CBECS ft^2/employee factors.

Calibration. Per ERCOT weather zone, $\text{ERCOT}_z(t) = \text{RES}_z\, r_z(t) + \text{COM}_z\, c_z(t) + \text{IND}_z$, fitted by non-negative least squares and scored by the same blocked seasonal CV. IND_z is a *fitted* estimate of each zone's weather-insensitive baseload, not an assumption.

2 Model accuracy

IECC zone	Residential mean R^2	Commercial mean R^2
2A (very hot-humid, incl. Coast)	0.801	0.916
3A (warm-humid, DFW/East)	0.642	0.864
3B (warm-dry, West TX/Permian)	0.479	0.426
2B (hot-dry)	0.456	0.742
4B (mixed-dry, Panhandle)	0.435	0.880

Table 1: Honest blocked-seasonal-CV mean R^2 by climate zone. Zone 2A, the Coast's own climate, is clearly easiest; drier and more continental zones are harder, consistent with a real heating season making the weather–load relationship less purely cooling-driven.

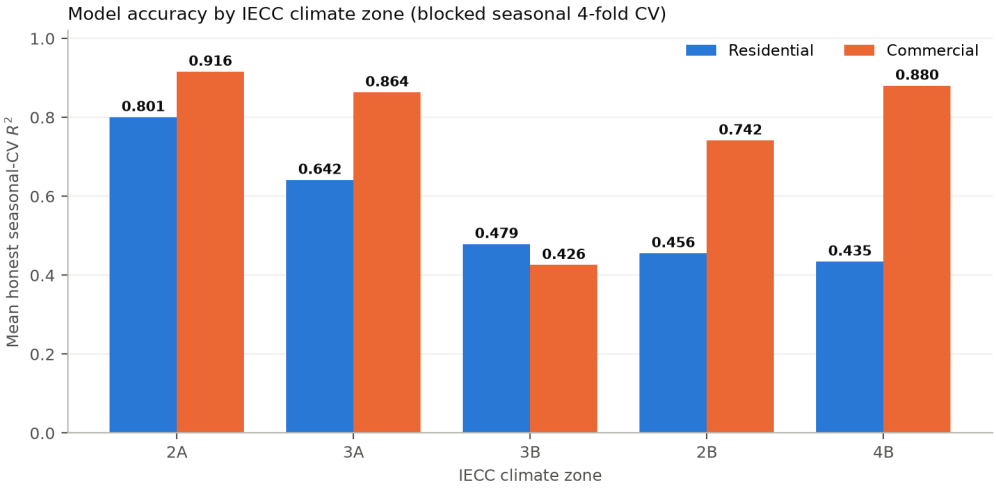


Figure 1: Model accuracy by climate zone. Commercial exceeds residential everywhere except zone 3B, where hotel and grocery archetypes carry genuinely no weather signal (Permian oilfield workforce cycles); four 3B/2B archetype combinations have negative mean R^2.

3 Spatial allocation: effect of the area split

	Nearest-bus join	Area-split
Residential, 50 parts per tract		
Tracts split across >1 bus	0%	84.0%
Buses receiving dwellings	3,426	**6,352**
Load-carrying buses with zero dwellings	1,824	**25**
Spearman ρ vs. bus `Load_MW`	0.502	**0.606**
Commercial, 100 parts per ZCTA		
ZCTAs split across >1 bus	0%	90.2%
Buses receiving floor space	1,598	**6,329**
Load-carrying buses with commercial load	20.1%	**96.1%**
Spearman ρ vs. bus `Load_MW`	0.133	**0.475**

Table 2: Scored against the Texas 7k case's own per-bus `Load_MW`. Spearman rather than Pearson: `Load_MW` also contains industrial demand, so a few very large industrial buses dominate any linear correlation.

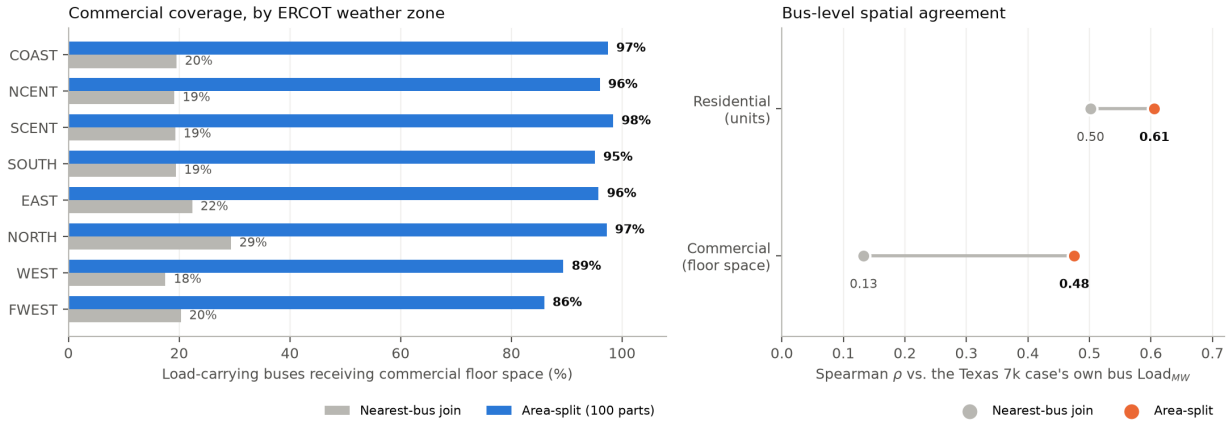


Figure 2: Left: share of load-carrying buses receiving commercial floor space, by ERCOT zone. Right: bus-level rank agreement with the case's own per-bus load.

Worth stating in the paper. This improvement is *invisible* to every zone-aggregate metric: redistributing load among buses of the same zone leaves that zone's total unchanged (zone totals moved by -1.6% to $+7.8\%$, most under 2%). Bus-level fidelity is what the framework exists to produce, so a validation protocol based only on aggregate correlation would miss it entirely — which is the argument for reporting the `Load_MW` rank statistic alongside the usual metrics.

4 Validation against ERCOT

4.1 Per-zone and statewide

ERCOT zone	ERCOT peak (MW)	Recon. peak (MW)	Peak ratio	LF (ERCOT)	LF (recon.)	Pearson r
COAST	20,269.8	19,613.6	0.97	0.600	0.620	0.900
NCENT	26,499.2	27,100.4	1.02	0.524	0.512	0.901
SCENT	14,166.7	12,650.0	0.89	0.484	0.542	0.902
SOUTH	6,176.1	5,734.7	0.93	0.577	0.621	0.911
WEST	2,083.6	1,919.7	0.92	0.612	0.664	0.823
FWEST	3,655.0	3,471.3	0.95	0.782	0.824	**0.655**
EAST†	2,620.8	2,700.1	—	0.569	0.552	0.911
NORTH†	1,521.8	1,415.5	—	0.571	0.614	0.905
STATEWIDE	**73,308.2**	**72,617.0**	**0.99**	**0.586**	**0.592**	**0.922**

Table 3: Calibrated reconstruction vs. ERCOT actual load, 2018. †The synthetic case's EAST and NORTH area boundaries do not coincide with ERCOT's (their case peak load is 1.33× and 2.21× the ERCOT zonal peak), so their shape statistics are interpretable but their magnitudes are not.

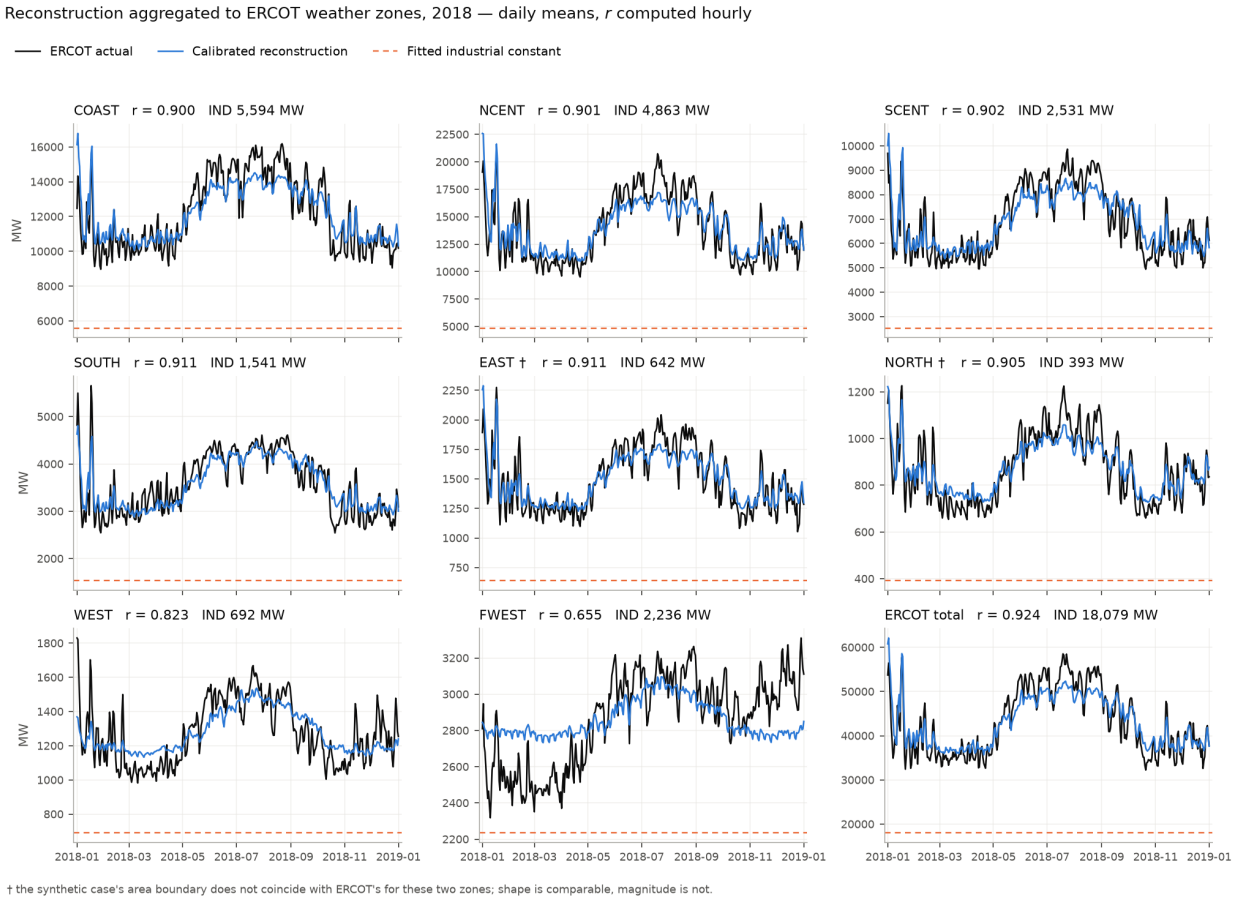
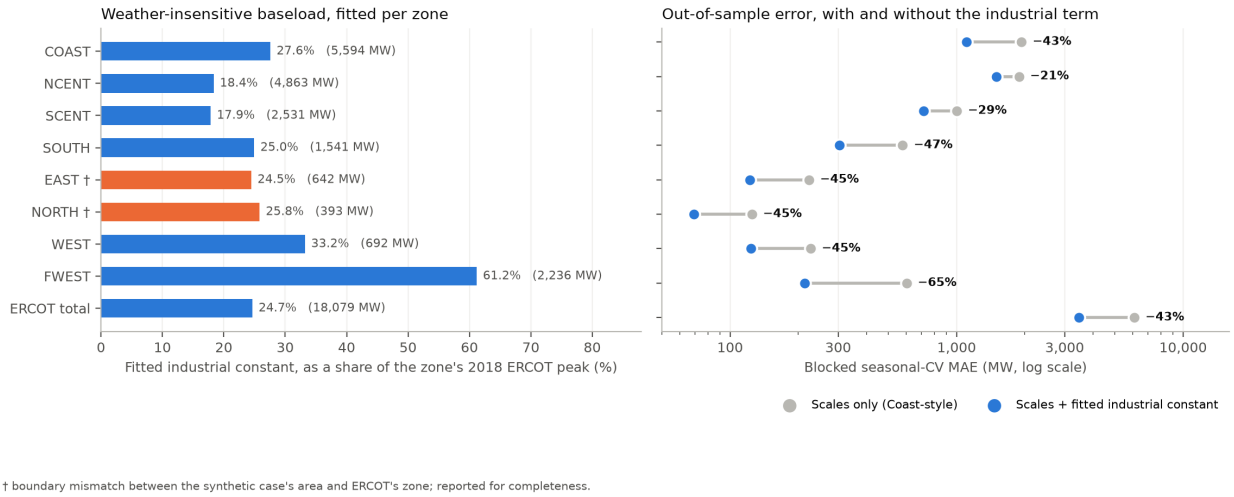


Figure 3: Calibrated reconstruction aggregated to each ERCOT weather zone. Daily means plotted for legibility; r computed hourly.

4.2 Calibration coefficients

ERCOT zone	RES	COM	IND (MW)	IND as % of zone peak
COAST	1.07	0.98	5,594	27.6
NCENT	1.36	0.98	4,863	18.4
SCENT	1.26	0.00*	2,531	17.9
SOUTH	1.14	0.66	1,541	25.0
WEST	1.07	3.03*	692	33.2
FWEST	**0.30**	1.46	2,236	**61.2**
EAST†	0.58	0.70	642	24.5
NORTH†	0.58	0.77	393	25.8
Statewide fit	**1.15**	**0.76**	**18,079**	**24.7**

Table 4: Per-zone calibration, NNLS, scored by blocked seasonal CV. *see Section 6.



† boundary mismatch between the synthetic case's area and ERCOT's zone; reported for completeness.

Figure 4: Left: fitted weather-insensitive baseload as a share of each zone's ERCOT peak. Right: out-of-sample error with and without the industrial term.

Three results to carry into the paper.

1. **The building-stock reconstruction needs almost no fudge factor.** With an industrial term present, RES sits at 1.07–1.36 in five of the six well-matched zones, against 1.6–1.9 for Coast without one. The earlier large scale factors were compensating for a missing *load component*, not for a mis-sized housing stock.

2. **Industrial demand is roughly a quarter of ERCOT** (18,079 MW statewide, 24.7% of system peak). Coast alone fits 5,594 MW, which retrospectively validates the hand-picked 4,500 MW placeholder as the right order of magnitude and slightly low.

3. **FWEST is structurally industrial**: RES = 0.30 and an industrial constant equal to 61.2% of its peak — a quantitative explanation for its weak correlation, and the clearest evidence that a weather-driven reconstruction cannot track Permian load.

5 Validation metrics beyond aggregate correlation

The metric suite requested in review point 5, computed statewide with the same definitions used for Coast.

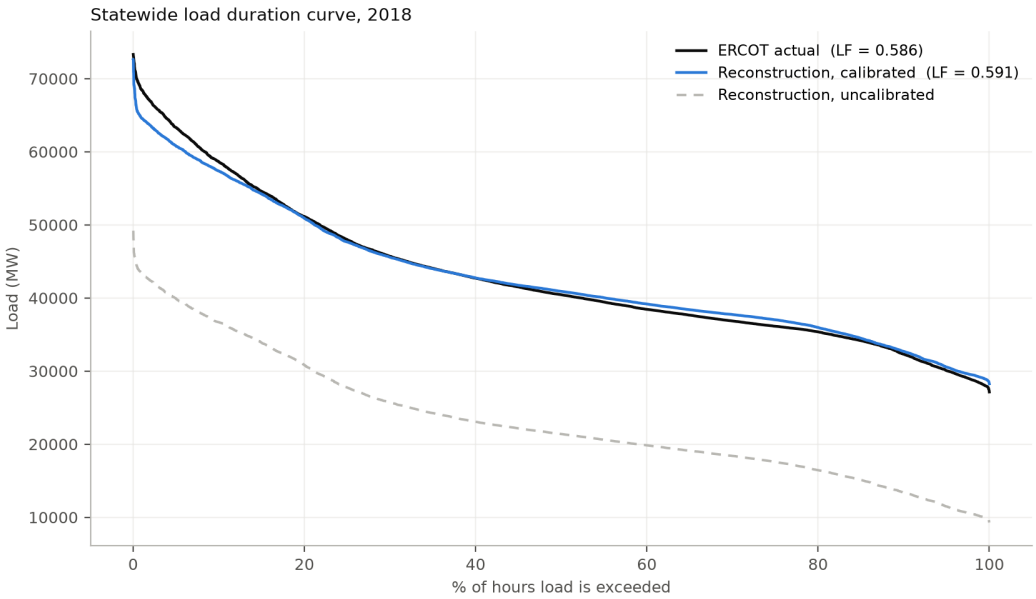


Figure 5: Load duration curve. The calibrated curve tracks ERCOT across the whole range; the load factor matches to within 0.006 (0.592 vs. 0.586), against 0.556 vs. 0.600 for the Coast study.

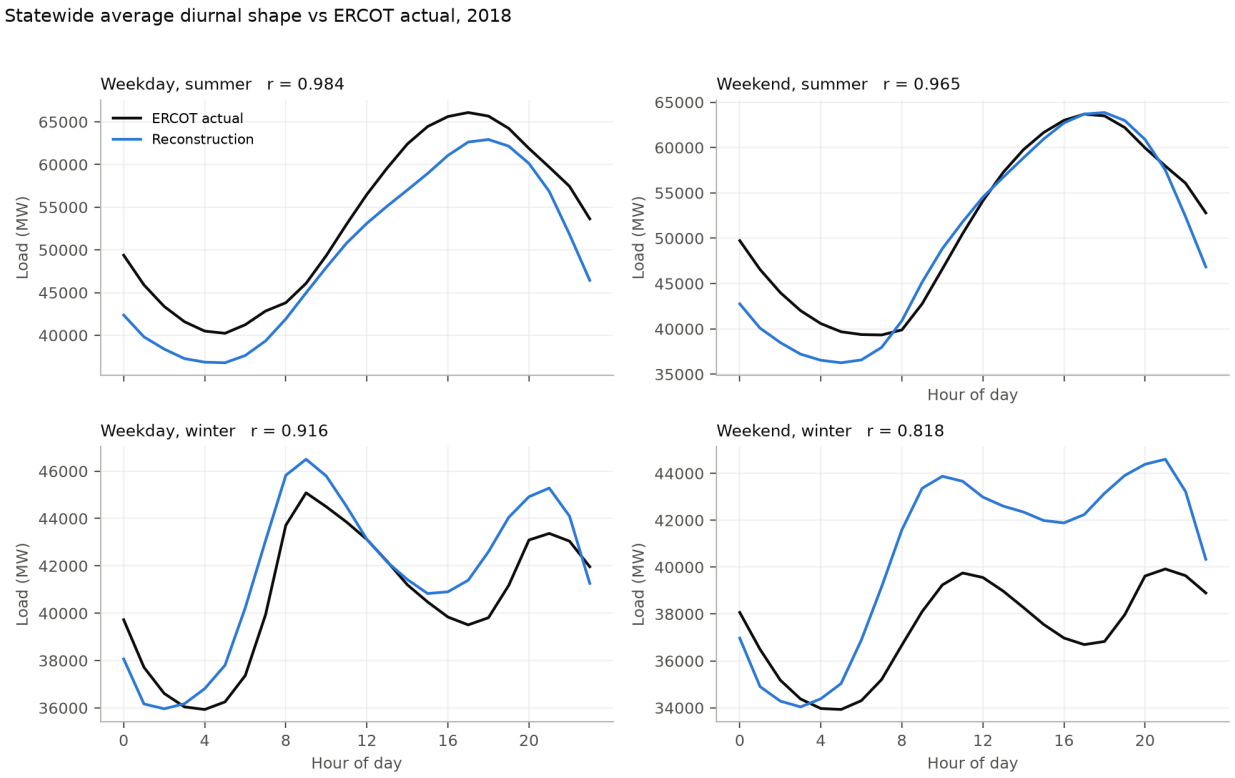


Figure 6: Average diurnal shape. Summer is reproduced almost exactly ($r = 0.984$ weekday, 0.965 weekend); winter weekend remains the weakest condition ($r = 0.818$), as it was for Coast.

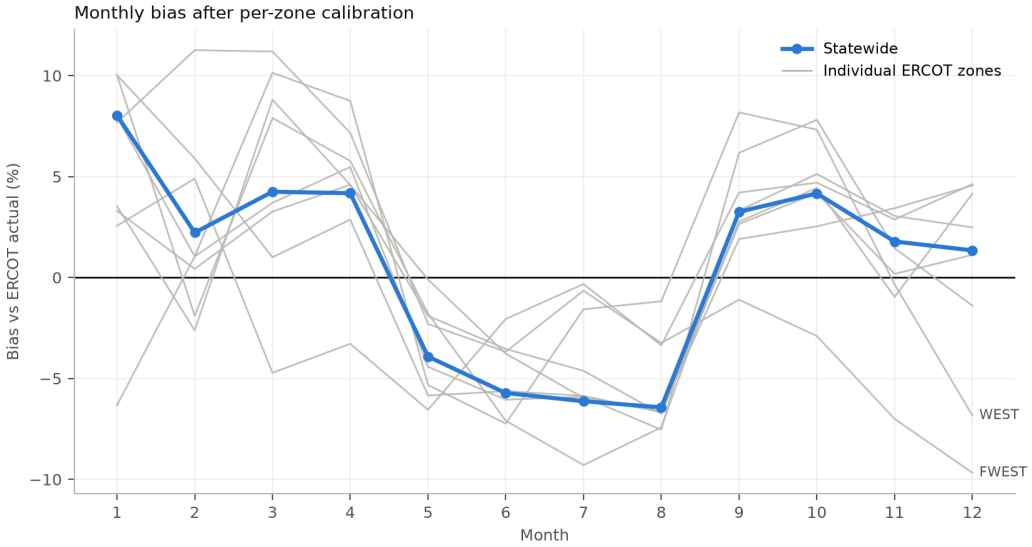


Figure 7: Monthly bias. Statewide bias stays within −6.4% (August) and +8.0% (January), mean absolute 4.3%, and changes sign across the year — a shape residual, not a scale error. The Coast study's bias was positive in every month and ranged +10.5% to +29.8%.

Scope	CF residential	CF commercial	CF combined
Statewide (6,717 buses)	**0.954**	**0.975**	**0.941**
COAST	1.000	0.997	0.990
NCENT	0.992	0.998	0.992
SCENT	0.981	n/a*	0.981
SOUTH	0.854	0.970	0.875
WEST	0.899	0.975	0.926
FWEST	0.993	0.998	0.978
EAST†	0.987	0.994	0.983
NORTH†	0.897	0.985	0.886

Table 5: Coincidence factor (system peak / sum of individual bus peaks); diversity factor is its reciprocal, so the statewide combined DF is 1.063. *undefined: SCENT's fitted COM = 0 makes the commercial stack identically zero.

The coincidence factor is finally informative. In the Coast study it sat at 0.96–1.00 across every pipeline version — a structural artifact of all 200 buses sharing one zone-wide weather series, not a physical result. Statewide, buses span eight weather zones and five climate zones, and the combined CF drops to **0.941**. Within a single zone it is still near 1 (COAST 0.990, NCENT 0.992), so the diversity being measured is *between* climate zones, not between buses. That is an honest reading, and it identifies per-bus weather granularity as the remaining lever.

Spatial consistency proxy	Coast (200 buses)	Statewide (6,717 buses)
Residential annual energy vs. dwellings	0.948	**0.973**
Commercial annual energy vs. floor space	0.892	**0.971**

Table 6: Correlation between per-bus annual energy and its per-bus driver. The commercial jump is the area split: under the nearest-ZCTA join most buses had zero floor space but non-zero modelled energy.

6 Limitations to state explicitly

1. **Industrial load is a constant, not a model.** It is fitted rather than assumed, which is an improvement, but it cannot reproduce industrial load's own temporal variation. This caps FWEST at $r = 0.655$ and WEST at 0.823. A time-varying, data-driven industrial representation is the single most consequential remaining gap.

2. **The area split assumes uniform density within a polygon.** Equal-area subdivision is a weak assumption in large rural tracts with one town, and nearly harmless in dense urban ones — which is where most load sits. It is still strictly better than assigning a whole polygon to one bus, and `Load_MW` now provides a per-bus reference to quantify the residual.

3. **EAST and NORTH have boundary mismatches.** The synthetic case's area load is 1.33× and 2.21× their ERCOT peak, so only their shapes are comparable.

4. **South Central fits** COM = 0 **and West fits 3.03.** For SCENT, the partial correlation of the commercial profile with ERCOT load (residential removed) is −0.014, against +0.24 to +0.34 in every other zone: the commercial stack genuinely adds no information there. Poor

coverage and collinearity were both tested and ruled out (coverage is now 96%; VIF ≈ 1.9). Unexplained, and isolated to the Austin/San Antonio corridor.

5. **All buses in a zone share one weather series.** Table 5 shows the consequence directly: within-zone coincidence factors remain near 1. Per-bus or per-county weather is the natural next refinement.

6. **Four archetype/zone combinations have negative R^2** (3B largehotel -1.36, 3B smallhotel -0.28, 2B smallhotel -0.26, 3B grocery -0.10). These are lumpy, low-signal archetypes in oilfield regions; worth excluding from calibration rather than letting them contribute noise.

Where the numbers live

Every figure here is in `latex_report/figures/`; every table is backed by a CSV in `statewide/metrics/`: `statewide_validation_summary.csv` (Table 3), `zone_calibration.csv` (Table 4), `statewide_coincidence_diversity.csv` (Table 5), `statewide_monthly_bias.csv`, `statewide_diurnal_corr.csv`, `statewide_spatial_proxy.csv`, and `seasonal_cv_{ZONE}.csv` plus `seasonal_cv_comstock_{ZONE}.csv` (Table 1). Reproduce with `statewide/notebooks/`: `00_zone_mapping` → `residential_{ZONE}` and `commercial_{ZONE}` → `zone_calibration` → `validation_metrics_statewide`.